

# Voltiq App Summary

Repository-based one-page summary

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| What it is   | Voltiq is a Next.js web app for renewable energy engineering workflows. It guides teams from early site screening through design checks, storage, finance, ESG analysis, and final report assembly. |
| Who it's for | Primary users are energy engineers and renewable project teams, especially solar developers, technical consultants, and C&I; decarbonization or energy-management teams.                            |

## What it does

- Groups work into four phases: Site & Resource, Technical Design, Storage, and Financial & ESG.
- Provides workflow templates for solar development, energy consulting, and C&I; optimization.
- Includes 15 engineering tools covering site scoring, land-use capacity, solar and wind yield, shading, PV loss, inverter sizing, cable sizing, storage, ROI, LCOE, carbon, Scope 2, and hydrogen.
- Pulls location and climate inputs from Nominatim and Open-Meteo, plus carbon signals from Electricity Maps where relevant.
- Saves tool snapshots and project metadata in browser localStorage for resume-able workspaces.
- Tracks phase progress, recent outputs, and recommended next actions inside the project workspace.
- Builds client-ready PDFs from saved outputs and can generate Gemini-based executive summaries.

## How it works

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|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UI shell        | Next.js Pages Router in `pages/`, global layout in `_app.jsx`, shared UI/layout components.                                                                                                            |
| Tool layer      | Route pages mount tool components in `components/tools/`; calculators live in `lib/*Calc.js`.                                                                                                          |
| Data flow       | Browser UI -> local calc and/or third-party fetch -> normalized snapshot -> localStorage.                                                                                                              |
| State/reporting | `reportStorage.js` and `projectWorkspace.js` manage saved outputs and workflow progress; `ReportGenerator.tsx` reads snapshots, optionally calls `lib/gemini.js`, then exports via `lib/pdfExport.js`. |
| Backend         | No in-repo API routes or database layer found in repo.                                                                                                                                                 |

## How to run

- Open the app folder: `cd voltiq`
- Set `NEXT\_PUBLIC\_GEMINI\_API\_KEY` in `.env.local` for Gemini features; the variable is present in repo config.
- Install dependencies: `npm install`
- Start dev mode: `npm run dev`
- Open the local URL printed by Next.js in the terminal.